C1 · bare original finger with a strain-sensing root



Original frame + TPU · fixed carrier · floating carrier · metal strain beam · two end clamps

CAD rendering. Gold markers locate active gauge grids; full gauge backings, leads and wrist electronics are not shown.

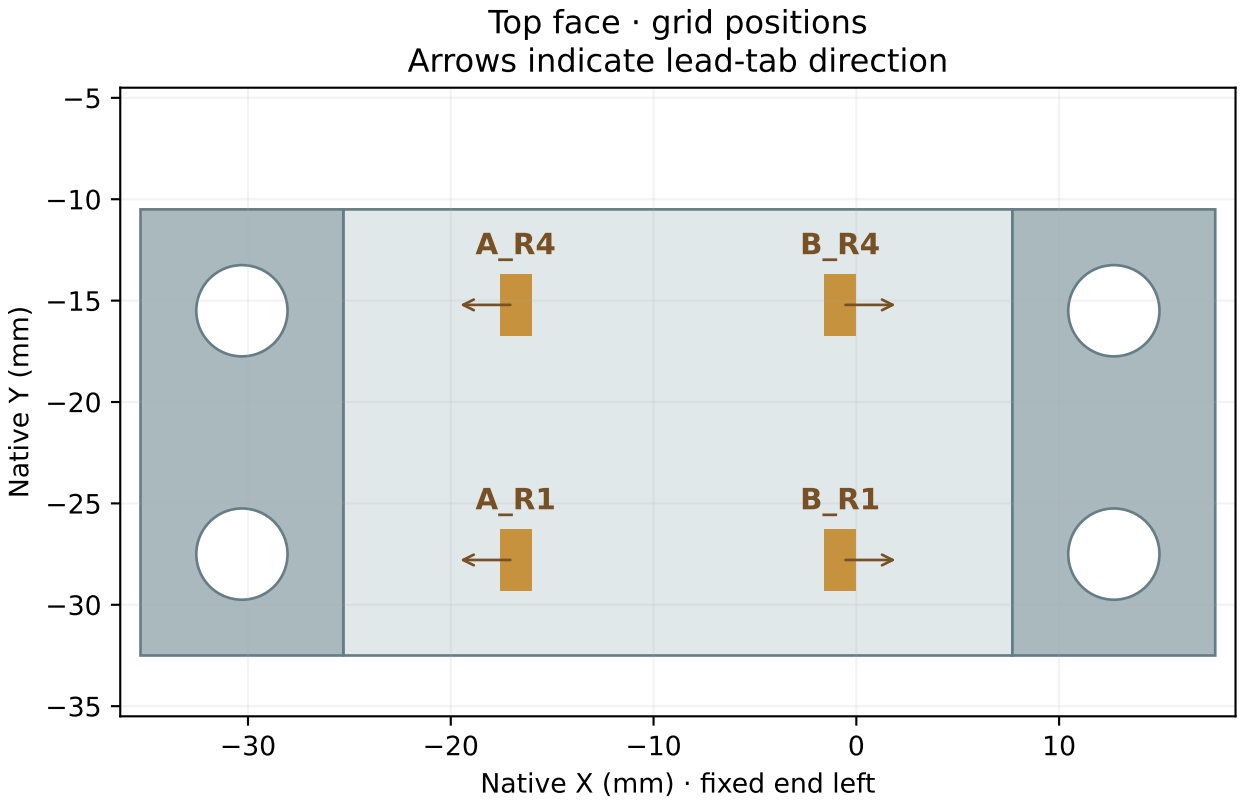
C1 · exploded root-force assembly

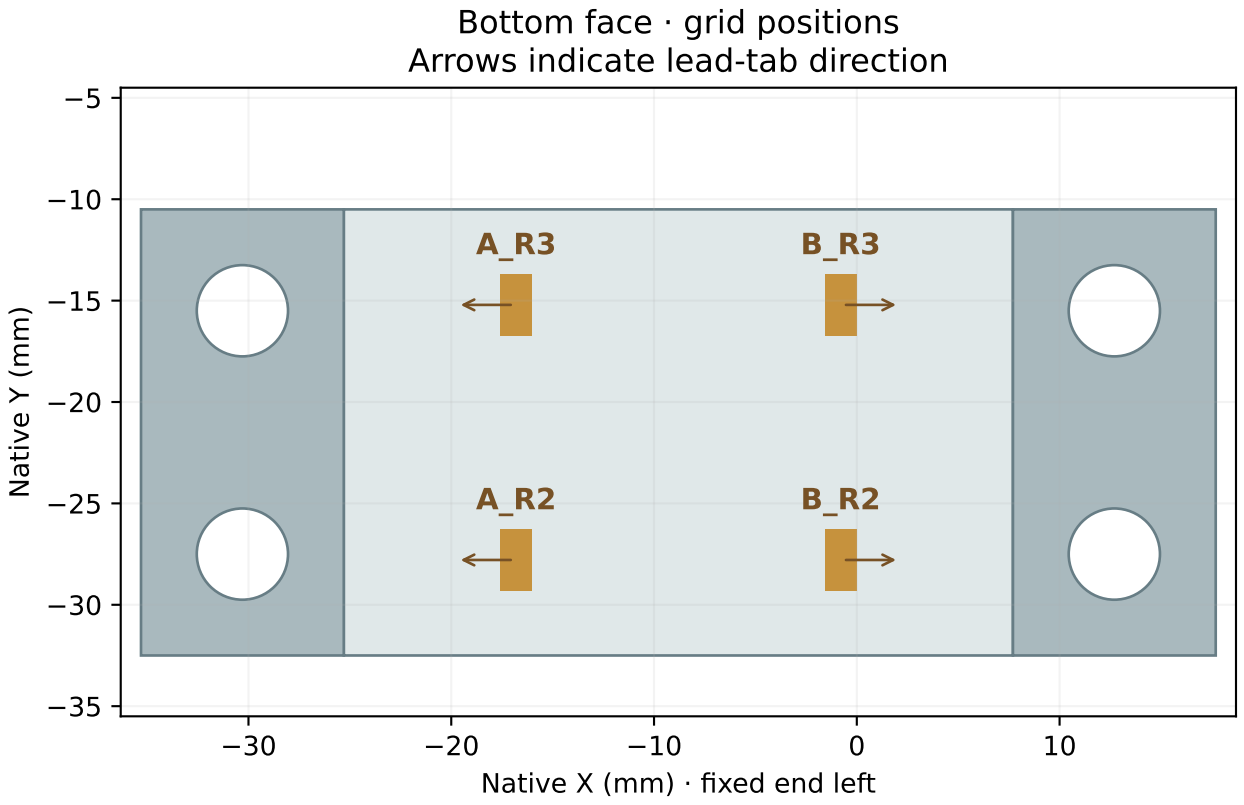


Original frame + TPU · fixed carrier · floating carrier · metal strain beam · two end clamps

CAD rendering. Gold markers locate active gauge grids; full gauge backings, leads and wrist electronics are not shown.

Eight CEA-13-062UWA-350 gauges · four per bridge · align the active grid, not the backing center





C1.1 manufacturing review · dimensions in millimeters · use STEP masters

Metal beam: 33 active + two 10 end clamps;
  width 22, thickness 14, 6061-T6 aluminum.
All load-path parts must be metal. Printed copies are fit witnesses.

Added reach 18.0; global width addition 21.40.
Added back height with reserved hardware: 24.22.
Original nail and TPU retained; no contact film is needed on the nail.

Gauge grid centers (X, Y, Z), with grid axes parallel to native X:
  A_R1: (-16.800, -27.790, 44.500)
  A_R2: (-16.800, -27.790, 30.500)
  A_R3: (-16.800, -15.210, 30.500)
  A_R4: (-16.800, -15.210, 44.500)
  B_R1: (-0.800, -27.790, 44.500)
  B_R2: (-0.800, -27.790, 30.500)
  B_R3: (-0.800, -15.210, 30.500)
  B_R4: (-0.800, -15.210, 44.500)

Grid: 1.57 x 3.05. Actual backing: 7.9 x 4.8; backing is asymmetric.
The larger keepout allows backing placement without guessing its grid offset.
Use the marking stencil and manufacturer bonding instructions.

Beam end through bores: 4.5. Carrier pilot bores: 3.3; tap M4 x 0.7.
Four M4 x 25 screws. Two 0.5 mm small-OD washers per screw; total stack 1.0.
Fixed clearance mount: M3 90-degree flat heads <=6.0 OD seated flush.
Alternative fixed carrier has 2.5 bores for M3 threads.
Floating rear hardware <=7 OD and <=4.5 projection; verify actual lengths.

Force range, drift, joint slip, endurance and robot tasks require physical tests.
Finite-element calculations use ideal bonded joints and do not qualify hardware.